Student ID _____ Student Name _____

University of the South Pacific

Faculty of Science, Technology and Environment

School of Computing, Information and Mathematical Sciences

IS221 Web Applications Development

Mid Semester Test

Semester 1, 2016

Version 1

Possible Marks:	**100**
Weight:	**20%**
Time Allowed:	**120** minutes
Number of Pages:	**17** (including this page)

Instructions:

1. Sections A, B, C, D : Answer ALL questions.

3. Write ALL answers in the spaces provided in the Answer Booklet.

- If you do need extra space for your answers, ask the lecturer.

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Section A: Multiple Choice 20 Marks

Instructions:
Identify and circle the choice that best completes the statement or answers the question. There is only **one** possible answer.

1) When uploading your website to be hosted on a remote server, _____ is the protocol being used to help you achieve this activity.
   a. HTTP/HTTPS
   b. SMTP
   c. UDP
   d. FTP

2) <p> and are examples of _____ elements.
   a. Deprecated
   b. Empty
   c. Inline
   d. Block

3) <img alt="company logo" src="myphoto.jpg" width="300" height="200">, the src attribute suggests that myphoto.jpg is _____.
   a. Located on a different server.
   b. Located in the same folder as the html document.
   c. Located in a different folder to that of the html document.
   d. Located in a different partition.

4) When trying to access your Facebook account, www.facebook.com corresponds to an IP address of a remote _____. Translation of web addresses to IP addresses is done by the _____.
   a. Database server, DNS
   b. Web server, DNS
   c. DNS, Database server
   d. DNS, Web server

5) Given the URL: http://www.usp.ac.fj/IS221/lab4.html, http is the _____ part of the URL.
   a. Domain Name
   b. Path Name
   c. Protocol
   d. File Name

6) To create a style that will be applied to a heading which is to be colored red and center aligned, the following style should be used:
   a. h1{ color: red; align:center; }
   b. h1{ color=" red"; align="center"; }
   c. h1{ color: red; align='center'; }
   d. h1{ color=' red'; align:right; }

7) World Wide Web Consortium sets standards for_____ while ICAAN is in charge of _____.
   a. HTML, domain names
   b. Domain names, XHTML
   c. Markup languages, domain names
   d. XHTML, domain names

8) Evolution of markup languages to what it is now is best summerised by:
   a. XML, HTML, XHTML, HTML5
   b. XHTML, HTML, XML, HTML5
   c. HTML, XML, XHTML, HTML5
   d. XML, HTML, XHTML, HTML5

9) To develop websites that are attractive, easy to navigate and compelling to use, web developers should consider:
   a. Crafting the look and feel, putting as much graphics as possible and use of active white space
   b. Crafting the look and feel, use 7 plus or minus 2 clicks to get to desired information
   c. Crafting the look and feel of the website specifically for just Google Chrome
   d. Crafting the look and feel, use graphics sparingly and observe the 3 click rule for information access.

10) Mary had just created an animated picture of a dancing island girl for her webpage. The file format that she should use to save this picture to ensure that the animation is saved is _____
   a. JPEG
   b. PNG
   c. GIF
   d. JPG

11) To create a table with a border width of 1, margin width of 1 between cells and cells that take up 2 rows, the following tags should be used:
   a. <table border = "1" >, <table cellpadding = "1" >, <table rowspan = "2" >
   b. <table border = "1" >, <table cellspacing = "1" >, <table colspan = "2" >
   c. <table border = "1" >, <table cellspacing = "1" >, <table rowspan = "2" >
   d. <table border = "1" >, <table cellpadding = "1" >, <table colspan = "2" >

12) _____ is a measurement unit used in _____ page layout such as that used by Digicel Fiji to ensure that the web designer has more control over how the website is rendered.
   a. percentage, flexible
   b. pixels, fixed
   c. percentage, fixed
   d. pixels, flexible

13) In a page template, the section that will normally contain the company's featured story would be _____.
    a. Page banner
    b. Copyright footer
    c. Content
    d. Navigation

14) The web development team member that would be predominantly responsible for creating the Web site's Specification will be _____:
    a. Designer
    b. Database Administrator
    c. Developer
    d. Applications Programmer

15) For a restaurant that takes customers' orders on its website which has a customized home page for the different logged in stakeholders such as customers and suppliers, the most appropriate content goal for this website is _____.
    a. Special interest
    b. Portal
    c. Publishing
    d. Intranet/extranet

16) Youtube has web analytics tool that allows subscribers who uploaded videos to view stats for each uploaded video. This tool facilitates _____ analysis.
    a. audience
    b. performance
    c. product
    d. code

17) Digicel's website uses both _____ and _____ site structures due to the extensive content of the website and the visibility of the main page links from any page on the website.
    a. Hierarchical, Web
    b. Web, Catalog
    c. Tutorial, Web
    d. Catalog, Tutorial

18) Maria, an orchid enthusiast would like to register a domain name for her site which will feature information about the different varieties of orchids and how to use them in the Japanese flower arrangement art of Ikebana. An appropriate domain name could be:
    a. www.orchidsbymaria.org.fj
    b. www.orchidsbymaria.com.fj
    c. www.orchidsbymaria.info.fj
    d. www.orchidsbymaria.net.fj

19) Three factors to take into consideration when deciding on a web hosting company for a website are:
    a. Technical support, SSL support and customer support
    b. Technical support, SSL support and number of email addresses provided
    c. Technical support, number of email addresses provided and customer support
    d. Technical support, Database support and customer support

20) Social media presence plays a crucial role in raising public awareness for a website. Creating social media presence for a site would be undertaken during the _____ phase of the web development lifecycle.
    a. Requirements analysis
    b. Design and template creation
    c. Implementation
    d. Publishing and marketing

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Section B: Matching Questions 20 Marks

Write the letter of the correct term that matches the given description in the space provided

A.	*HTML5*	**1.**	_____ Strict, Transitional and Frameset
B.	*<table>*	**2.**	_____ Protocol used during online banking.
C.	*Server farms*	**3.**	_____Separation of presentation properties from content
D.	*XHTML*	**4.**	_____ Used within to structure information
E.	*Browser cache*	**5.**	_____ Interprets HTML tags and renders the web page
F.	*Style declaration*	**6.**	_____ Temporary storage for web graphics
G.	*HTTPS*	**7.**	_____ Used by Google and Facebook to handle web requests
H.	*<th>*	**8.**	_____ Logical element that renders text bold
I.	*Validator*	**9.**	_____ Checks for well-formed XHTML documents.
J.	*Compatibility*	**10.**	_____ Defines a table header cell.
K.	*Rendering engine*	**11.**	_____ Reserved for domain names affiliated with the military
L		**12**	_____ Website is accessible on Chrome, IE, Safari and Firefox
M	*.mil.fj*	**13**	_____ Knowledge pertaining to networks, protocols and Apache hardware
N	*Portability*	**14**	_____ Shopping cart and search functionality mostly at top right corner of web page
O	*Server administrator*	**15**	_____ Consists of a selector and a declaration
P	*Designing for location*	**16**	_____ File naming standard

Q *Operability*

R *WCAG*

S *Multiple resolutions*

T. *ISO9960*

17 _____ Being able to use the web top up feature of Digicel to successfully top up your phone

18 _____ Website is accessible on IPhone, IPad, Android, Mac and Windows desktops

19 _____ Fixed and flexible page layouts

20 _____ Avoid using flashing content on a website as this could trigger seizures

Section C: Coding Questions 30 Marks

1) Given the source code below, with the help of a diagram illustrate how the web page is going to be rendered. (15 Marks)

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Europa</title>
<style type="text/css">
    .style3
    {
        color: gold;
        text-align: center;
    }
    .style4
    {
        color: gold;
        font-family: "AR BERKLEY";
        font-size: xx-large;
    }
    .style7
    {
        color: gold;
        text-align: center;
        text-decoration: underline;
    }
    .style8
    {
        width: 100%;
    }
    </style>
</head>
<body>

<table border="0" bgcolor="black" style="width: 100%">
<tr>
<td width="17%" height="120" >
<img alt="Europa logo" width="100%" height="150"
        src="Images/Europa_logo.PNG" />
</td>
<td align="center" width="83%">
 <img alt="" height="150" src="Images/Europa_Mission_statement.PNG"
        width="100%" /></td>
</tr>
</table>
<table width="100%"border="0" bgcolor="black">
<tr>
<td width="15%" class="style3"><a href="http://www.europa.co.fj/home.html">Home</a></td>
<td width ="15%" class="style3"><a href="http://www.europa.co.fj/about_us.html">About
Us</a></td>
<td width = "15%" class="style3">
    <a href="http://www.europa.co.fj/registration.html">Registration</a></td>
<td width = "15%" class="style3">
    <a href="https://www.europa.co.fj/reservations.html">Reservation</a></td>
<td width = "15%" class="style3"><a
href="http://www.europa.co.fj/gallery.html">Gallery</a></td>
<td width = "15%" class="style3"><a
href="http://www.europa.co.fj/contact_us.html">Contact Us</a></td>
```

```
</tr>
</table>
<table border="1"  width="100%" cellspacing="0" cellpadding="0">
<tr>
<td width="17%" bgcolor="black" rowspan="2">
<p align="center">
    <br/>
    <a href="http://www.europa.co.fj/french_cuisine.html">
    <img alt="" height="50" src="Images/French_Cuisine1.jpg" width="90" style="border-
width: 0px" /></a><p align="center" class="style7”>
        French Cuisine<br/><br/>
    <a href="http://www.europa.co.fj/german_cuisine.html">
    <img alt="" height="50" src="Images/German_Cuisine.jpg" width="90"
        style="align:center" style="border-width: 0px"/></a><p class="style7">
        German Cuisine<br/><br/>
    <img alt="" height="50" src="Images/Italian_Cuisine.jpg" width="90"
        style="text-align: center" /><p align="center">
        <span class="style5">Italian Cuisine</span><br/><br/>
    <img alt="" height="50" src="Images/Spanish_cuisine.jpe" width="90" /><p
        class="style7">Spanish Cuisine</p>
    <p
        class="style7">
        <a href="http://www.facebook.com/europafiji">
        <img alt="" height="50" src="Images/like_us_on_facebook.jpe" width="90"
            style="border-width: 0px" /></a></p>
</td>
<td bgcolor="black" valign="top" colspan="2" >
<h1 align="center" class="style4" >Chef&#39;s Specials....</h1>
    <table class="style8" align="center" cellpadding="2" cellspacing="2">
        <tr>
            <td>
                <img alt="French dish made of snails" height="200"
src="Images/Escargot.jpg" width="300" /></td>
            <td>
                <img alt="Italian dish made of pasta and chicken"
src="Images/chicken%20pasta%20salad.png" height="200" width="300" /> </td>
            <td>
                <img alt="Spanish dish made of rice, prawns and mussels"
src="Images/spanish%20paella.jpg" height="200" width="300" /> </td>
        </tr>
        <tr>
            <td table colspan = 3 style="text-align: center">
                <span class="style4">
                <br />
                For that special occasion where divine European
        cuisine is what your heart desires... Europa is the place to be...</span></td>

        </tr>
    </table>
</td>
</table>
</body>
</html>
```

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2) Given the source code below identify five specific errors, state the XHTML syntax rule that the error violates and state how to correct it using examples. (15 Marks)

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
    "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html>

<head>
<title> IS221 – MST Test Information </title>

         <style type="text/css">
        .style1
            {
                color: #0033CC;
                text-align: center;
            }
        </style>

</head>

<body>
<div>
<img src="images/usp_logo.jpg" alt = "usp logo" >

<h2 class="style1"> IS221 S1 2016 MST Information</h2>

<table border = "0" width = "90%">
<tr>
<td width = "100 pixels">
<a href="aboutme.html"><img src="images/aboutme.gif" alt="About Me" border = "0" /> About
Me</a>
<Br>
<Br>
<a href="http://www.usp.ac.fj"> <img src="images/home.png" alt="" border = "0"/> USP </a>
<Br>
<Br>
<a href="Activity2.html"> <img src="images/xhtml.gif" border = "0" /> XHTML </a>

</td>

<td>
<table align=center>
<tr>
<td width = "200 pixels"> Multiple Choice Questions:
</td>
<td width = "80 pixels"> 20</td>

</tr>

<tr>
<td width = "200 pixels"> Short Answer Questions:
</td>
<td width = "80 pixels"> 3
</td>

</tr>

<tr>
<td width = "200 pixels"> Coding Questions:
</td>
<td width = "80 pixels"> 2</td>
```

```
</tr>


</table>
<h2 class="style1"> MST  Coverage</h2>
<table border ="1 pixel" width = "400 pixels" align="center">
<tr>
<th> Readings </th>
<th> Lectures </th>
<th> Labs</th>

</tr>
<tr>
<td>Reading 1 - 5 </td>
<td>Lectures Wk1 - wk 5</td>
<td> Labs 1 - 3</td>
</tr>

</table>

<h3 class="style1"> MST Questions </h3>
A breakdown of the MST questions are as follows:
<ol>
     <li>Section A

      <ul>
       <li>20 Multiple Choice questions</li>
       <li>1 mark each</li>
       </li> </ul>

       <li>Section B

        <ul>
         <li>20 Matching questions</li>
         <li>1 mark each
         </ul> </li>

       <li>Section C

        <ul>
           <li>2 Coding questions</li>
           <li>15 mark each</li>
           </ul> </li>


       </td>

       </tr>
       <tr>
       <td width = 100 pixels>
                     </td>

       <td>
        </td>

       </tr>
       </table>
       <hr/>

       </div>
       <div>
       <table align=center>
       <tr>
```

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```
<td><a href="aboutIS221.html"><img src="images/aboutIS221.gif" alt="About IS221" border =
"0" /> About IS221</a>  </td>
<td><a href="http://www.usp.ac.fj"> <img src="images/home.png" alt="" border = "0"/> USP
</a> </td>
<td><a href="Activity2.html"> <img src="images/xhtml.gif" border = "0" /> XHTML </a>
</td>
</tr>
</div>
</table>
<hr/>

</body>
```

Error 1_____

Syntax rule violated _____

Correction_____

Error 2_____

Syntax rule violated _____

Correction_____

Error 3_____

Syntax rule violated _____

Correction_____

Error 4_____

Syntax rule violated _____

Correction_____

Error 5_____

Syntax rule violated _____

Correction_____

Section D: Short Answer Questions 30 Marks

1. Digicel and Vodafone Fiji are two popular websites amongst mobile phone users. In designing these websites, the designers took into consideration the following pointers:
    - Multiple screen resolutions
    - Crafting the look and feel
    - Easy access to your information
    - Planning for Clear presentation of information
    - Design for interaction

For each consideration compare and contrast how Digicel and Vodafone websites implemented them.

(10 Marks)

Multiple screen resolutions (2 marks)

Crafting the look and feel (2 marks)

Easy access to your information (2 marks)

Planning for Clear presentation of information (2 marks)

2. One way to design the layout of a webpage is to use tables. With the aid of a diagram, illustrate how you would create a page template for a company that sells fruits and vegetables that has the following requirements for their home page:
    - Logo and mission statement should be featured
    - Links to the other pages of the website should be available
    - Fruit and vegetable specials should be a main feature content
    - Advertisements of fruit and vegetable related products, deals and news should also feature
    - Copyright statement should feature as well

(10 Marks)

3. You have been assigned the task of designing the home page for the fruit and vegetable company in question 2. Using the Web development life cycle main phases, discus how you would implement each of the phases.

(10 marks)

Requirements and specifications

_____
_____
_____
_____

Information design and page template creation

_____
_____
_____
_____

Construction and quality assurance

_____
_____
_____
_____

Publishing and promotion

_____
_____
_____
_____

Student ID _____ Student Name _____

Maintenance

_____
_____
_____
_____